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PROPOSITION DE STAGE EN COURS D'ETUDES

Référence : **Votre DDS-Année-Numéro d'ordre**
(à rappeler dans toute correspondance)

Lieu : Palaiseau

Département/Dir./Serv. : DTIS

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DESCRIPTION DU STAGE

Thématique(s) : IAD

Type de stage : ☒ Fin d'études bac+5 ☒ Master 2 ☐ Bac+2 à bac+4 ☐ Autres

Intitulé: Comparison of LLM-based information extraction methods for safety and certification

Sujet : Aircraft certification is a multi-step process aiming at proving that aviation products are compliant with strict airworthiness and environmental standards and regulations. The overall process is based on formal modeling and design and it is supported by information extraction techniques that are able to identify relevant information items within large collections of documents (standards, design and engineering specifications, etc.).

The objective of this work is to investigate the potential of Large Language Models (LLMs), in extracting information from text by using a realistic use case of aircraft certification. Recent studies illustrated the capabilities of LLMs to facilitate both information extraction and modelling, as they provide useful results for various tasks, such as: detection of semantic relations [2], structuring [6] or identification of regulatory requirements [3], and extraction of rules from natural language specifications [4]. More complex applications, aiming at building knowledge graphs or ontologies [5] can also rely upon LLMs, although guidance in the form of expert knowledge is needed [1].

For this work, information extraction aims at extracting concepts and relations that are relevant for aircraft certification from a collection of documents. LLMs will be used with various prompting strategies implementing distinct queries for information retrieval (for example specific queries to retrieve concepts or relations). In addition, traditional methods for information extraction will be used in order to provide a baseline for results evaluation and comparison.

Experiments will be implemented by following the steps:

1. Prompt engineering for information extraction (definition of strategies).
2. Building the evaluation baseline thanks to traditional information extraction techniques;
3. Comparison and revelation of results.

The work has the following milestones:

- State of art and problem analysis (1 month)
- Prompt engineering (1 month)
- Construction of the baseline (1 month)
- Experiments, analysis and evaluation of results (1,5 month)
- Internship report (0.5 month)

References:

1. Liu, X., Li, Z., He, M., Ma, Z., Wu, X., Yilmaz, G., ... & Jansson, P. (2026). From Prompt to Graph: Comparing LLM-Based Information Extraction Strategies in Domain-Specific Ontology Development. *Procedia Computer Science*, *277*, 50-60.

2. Hannah, G., de Berardinis, J., Payne, T. R., Tamma, V., Mitchell, A., Piercy, E., ... & Konev, B. (2025). Relrae: LLM-based relationship extraction, labelling, refinement, and evaluation. *arXiv preprint arXiv:2507.03829*.

3. Abualhaija, S., Ceci, M., Sannier, N., Bianculli, D., Lannier, S., Siclari, M., ... & Tosza, S. (2025, September). LLM-assisted extraction of regulatory requirements: A case study on the GDPR. In *2025 IEEE 33rd International Requirements Engineering Conference (RE)* (pp. 142-154). IEEE.

4. Liu, C., Wang, Z., Fu, D., & Lei, H. (2026, May). LLM-Driven End-to-End Compliance Strategy: From Natural Language Specifications to Executable Rules. In *2026 6th International Symposium on Computer Technology and Information Science (ISCTIS)* (pp. 329-337). IEEE.

5. Bakker, R. M., Di Scala, D. L., & de Boer, M. H. (2024, September). Ontology Learning from Text: an Analysis on LLM Performance. In *NLP4KGC@ SEMANTiCS* (pp. 70-87).

6. Norheim, J. J., & Rebentisch, E. (2024, June). Structuring natural language requirements with large language models. In *2024 IEEE 32nd International Requirements Engineering Conference Workshops (REW)* (pp. 68-71). IEEE.

Est-il possible d'envisager un travail en binôme ? Non

Méthodes à mettre en oeuvre :

☒ Recherche théorique	☒ Travail de synthèse
☒ Recherche appliquée	☒ Travail de documentation
☐ Recherche expérimentale	☒ Participation à une réalisation

Possibilité de prolongation en thèse : Non

Durée du stage : Minimum : 4 Maximum : 6

Période souhaitée : Février Aout 2027

PROFIL DU STAGIAIRE

Connaissances et niveau requis :	Ecoles ou établissements souhaités :
Programmation Python, RAG, LLM, connaissances en ingénierie des connaissances (OWL, RDF) ainsi que traitement du langage naturel serait un avantage.	Master 2, Fin d'études BAC +5